

PHAROS

A Physics-Hybrid, Attention-Routed Mixture-of-Experts Architecture for RNA Structure and Contact-Map Prediction

Architecture Design Report — Version 0.1

Shuvam Banerji Seal
SBS-RNA Project

12 September 2026

Abstract

RNA three-dimensional structure prediction has not had its AlphaFold2 moment. The best current methods reach a mean TM-score of approximately 0.55 on non-redundant benchmarks, which is the threshold for recovering roughly the correct fold, against routine scores above 0.9 for proteins. This report argues that the bottleneck is not model scale but *inductive bias*, and specifies an architecture, **PHAROS**, built around that claim.

Three empirical findings drive the design. First, ERNIE-RNA (86M parameters) outperforms RiNALMo (650M) and UNI-RNA (400M) on secondary-structure prediction, and nearly doubles cross-family generalisation relative to RiNALMo ($F_1 = 0.646$ versus 0.355), on the strength of a single hand-set, three-valued base-pairing prior added to its attention logits. Second, we measure on 117 non-redundant RNA chains that contacts per nucleotide *saturates* at 4.4–4.9 while contact-map density falls from 6.5% at $L \approx 56$ to 0.353% at $L \approx 2900$: RNA contact maps scale as $O(L)$, not $O(L^2)$, so a dense pair representation spends over 99% of its compute on empty space and would require 309 GB of activations at $L = 2048$. Third, we measure a monotonic 1.76σ gradient in normalised crystallographic B-factor as a function of distance to the nearest Mg^{2+} ion across 24,623 nucleotides, establishing that ion coordination and local rigidity are not independent features but two readouts of one phenomenon.

PHAROS accordingly (i) accepts **ionic conditions as a model input**, which no existing RNA structure predictor does, propagating them to attention logits through a Manning-renormalised screened-Coulomb bias—though an audit reported here shows the *learned* part of that claim cannot be trained on the PDB, whose recorded Mg^{2+} spans only 5–15 mM because nobody deposits unfolded RNA; (ii) replaces the dense pair track with a **hierarchical coarse-to-fine pair track** that selects contact-bearing *blocks* rather than pairs, at 2.2% of dense cost—adopted after we measured a flat top- K design recovering only *one contact in five* on long chains, and measured block occupancy falling to 1.34% as chains lengthen; (iii) routes a **sparse mixture-of-experts** on local structural regime rather than sequence context, supported by our negative result that GNRA k -mer context predicts rigidity at only 0.073σ ; and (iv) supplies motif geometry through a **frozen retrieval bank** of 667 RNA 3D Motif Atlas classes rather than expecting the model to learn it from 6,661 structurally-resolved sequences. We also identify two supervision channels— Mg^{2+} sites and B-factor rigidity—that are already present in every mmCIF file the field downloads and are currently used by no structure predictor.

This is a design specification. No model has been trained. All quantitative claims are either marked **[measured]** and reproducible from the scripts in this repository, or cited to the literature.

Contents

1	Problem statement	3
1.1	The accuracy gap	3
1.2	Why RNA lags	3

2	The central argument: bias beats scale	3
3	Empirical analysis	4
3.1	RNA contact maps scale linearly, not quadratically	4
3.2	Ion coordination and local rigidity are one phenomenon	5
3.3	Ion inventory and coordination chemistry	5
3.4	Coevolution is strong, but gated by alignment depth	6
3.5	A negative result that constrains the design	7
4	The PHAROS architecture	7
4.1	Inputs	7
4.2	Token trunk: hybrid attention with MoE	8
4.2.1	Physics-biased attention	10
4.2.2	Mixture-of-experts feed-forward	10
4.2.3	Parameter budget	10
4.3	Hierarchical pair track	12
4.3.1	A flat top- K proposal was specified, measured, and discarded	12
4.3.2	Contacts are clustered, so select blocks rather than pairs	12
4.3.3	Three-level coarse-to-fine track	14
4.3.4	Reference implementation and measured cost	14
4.4	Motif retrieval bank	15
4.5	Physics module	15
4.6	Heads and decoder	16
5	Ionic-condition supervision: an audit that rescoped the claim	16
5.1	Coverage is roughly one third, split across two storage conventions	16
5.2	The deeper problem: the PDB is survivorship-biased on ionic conditions	17
5.3	How the claim is rescoped	17
6	Data mapping and the acquisition gap	17
7	Novelty claims	19
8	Risks and open questions	19
9	Status	20
A	Reproducing the measurements	20

1 Problem statement

1.1 The accuracy gap

Table 1 gives the state of the art on a non-redundant RNA test set as of November 2025.

Table 1: Current state of the art in RNA 3D structure prediction.

Method	Mean TM-score	Approach
trRosettaRNA	0.548	MSA $\rightarrow$ restraints $\rightarrow$ folding
NuFold	0.525	AlphaFold2-style nucleotide frames
AlphaFold3 (server)	0.510	all-atom diffusion
AlphaFold3 (local)	0.504	all-atom diffusion

A TM-score near 0.5 means the topology is approximately right and little else. The CASP16 signal is equally instructive: leading RNA groups won by integrating AlphaFold3 *into physics-based modelling* rather than by scaling a neural network. Physics-plus-machine-learning hybrids are currently ahead in RNA, which is the premise this architecture is built on.

1.2 Why RNA lags

1. **Data scarcity.** Our catalogue records 6,661 unique sequences with experimental 3D structure against 1.37 billion sequences, a coverage of 0.0005%. Proteins had roughly 220,000 structures at the AlphaFold2 moment.
2. **Redundancy within that data.** 15,441 RNA3DB chains collapse to 3,157 unique sequences; 5S rRNA alone accounts for 5,976 BGSU equivalence classes.
3. **Family coverage.** Only 100 of 4,227 Rfam families (2.4%) have any 3D representative.
4. **Conformational plasticity.** Riboswitches are *defined* by adopting multiple functional folds; single-structure prediction is the wrong target for a large fraction of functional RNA.
5. **Ion dependence.** RNA is a polyanion. Its fold is not defined without specifying the ionic environment, yet no mainstream predictor accepts ionic conditions as an input.
6. **Target noise.** Median resolution in our catalogue is 3.10 Å, with only 13% of chains at 2.5 Å or better.

2 The central argument: bias beats scale

Table 2 contains the most important result in the recent RNA language-model literature.

Table 2: ERNIE-RNA versus larger dense models. A $7.5\times$ parameter deficit is overturned by one hand-set prior.

Model	Params	bpRNA-1m F_1	bpRNA-new F_1
ERNIE-RNA (attention map)	86M	0.873	—
ERNIE-RNA (embedding)	86M	0.862	0.646
RiNALMo	650M	0.850	0.355
UNI-RNA	400M	0.821	—
RNA-FM	100M	0.808	—

ERNIE-RNA’s entire architectural advantage is an additive bias on attention logits. In its first layer,

$$\text{Attention}(Q, K) = \frac{QK^\top}{\sqrt{d}} + B_{\text{pair}}(i, j), \quad \text{Attention}_l(Q, K) = \frac{QK^\top}{\sqrt{d}} + \text{Attention}_{l-1}(Q, K), \quad (1)$$

with B_{pair} set by hand to 3 for C–G, 2 for A–U, a tunable $\alpha \approx 0.8$ for G–U, and 0 on the diagonal.

The cross-family result is the decisive one: 0.646 versus 0.355. A fixed physical prior nearly doubles generalisation to unseen families, because the prior is true regardless of family whereas learned family statistics are not.

Thesis. If a crude three-valued prior buys more than $7\times$ the parameters, the marginal return on *better physics* exceeds the marginal return on *more scale*. PHAROS replaces that hand-set prior with a learned, ion-aware, motif-aware, coevolution-aware one.

Its limitations define our openings. The prior is static, so a G–U pair inside a kink-turn is treated identically to one in a free helix. It is nearest-neighbour-blind, whereas real pairing energy is stacking-dependent. It encodes only canonical pairing, while non-canonical pairs define RNA 3D motifs and carry most tertiary information. And it ignores ions entirely.

3 Empirical analysis

All results in this section were measured for this report on 180 BGSU non-redundant representative structures at ≤ 3.0 Å, downloaded and analysed by scripts in `.../sampling/`. Raw outputs are in `data/samples/analysis/`.

3.1 RNA contact maps scale linearly, not quadratically

Contacts are defined as nucleotide pairs with any heavy-atom pair within 8.0 Å and sequence separation $|i - j| \geq 4$.

Table 3: Contact density by chain length across 117 chains. **[measured]**

Chain length	n	median L	contacts/nt	map density
32–100	38	56	1.784	6.523%
100–200	6	124	2.851	4.043%
200–500	3	393	4.216	2.151%
500–1500	9	1038	4.532	0.974%
≥ 1500	61	2901	4.854	0.353%
Overall			median 4.397, p95 5.422, max 5.497	median 0.389%

Contacts per nucleotide saturates at 4.4–4.9 and is bounded by RNA coordination geometry, independent of length; density falls approximately as $1/L$. The number of contacts therefore grows *linearly* in L .

The consequence for architecture is immediate. A dense pair representation of dimension $c = 128$ requires, for 48 blocks holding roughly six intermediate tensors each and without gradient checkpointing:

$$L = 512 \rightarrow 19.3 \text{ GB}, \quad L = 1024 \rightarrow 77.3 \text{ GB}, \quad L = 2048 \rightarrow 309.2 \text{ GB}.$$

A dense pair track is not merely wasteful at RNA lengths; it is infeasible. How that sparsity is exploited turns out to matter as much as exploiting it at all: a flat top- K selection fails on long chains, and contacts must be selected as *blocks*. Section 4.3 reports both measurements.

3.2 Ion coordination and local rigidity are one phenomenon

We z-normalise crystallographic B-factors within each structure so that refinement protocols are comparable, and restrict this analysis to the 31 X-ray structures among the sample; cryo-EM atomic displacement parameters are not comparable and were excluded. Lower z_B indicates greater rigidity.

Table 4: Normalised B-factor versus distance to the nearest Mg^{2+} , over 24,623 nucleotides. **[measured]**

Distance to nearest Mg^{2+}	n nucleotides	mean z_B
0–4 Å	319	− 0.883
4–6 Å	2,441	−0.673
6–8 Å	7,590	−0.415
8–12 Å	8,115	+0.004
12–20 Å	2,984	+0.756
> 20 Å	3,174	+ 0.877

The gradient is monotonic and spans 1.76 standard deviations. Local packing density independently correlates with rigidity at Pearson $r = -0.294$.

Architectural consequence. Ion effects and motif rigidity, which the project brief treats as two separate physical ingredients, are strongly coupled. A model that predicts Mg^{2+} sites obtains a rigidity map nearly for free and vice versa. That mutual information justifies routing them into a *shared* physics expert rather than two disconnected heads.

Caveats. B-factor also absorbs resolution, occupancy and refinement choices. Mg^{2+} proximity and dense packing are themselves correlated, since both mark folded cores, so this is association rather than isolated causation; the gradient partly restates that structured cores both bind ions and are ordered. The closest bin contains only 319 nucleotides. This is strong evidence for *using* the coupling as a prior, not a proof of mechanism.

3.3 Ion inventory and coordination chemistry

Table 5: Cation inventory across 179 parsed structures. **[measured]**

Ion	Count	Inner-sphere partner	Count
Mg^{2+}	17,428	OP2 (phosphate)	5,890
K^+	1,868	OP1 (phosphate)	5,144
Zn^{2+}	359	O6 (guanine)	757
Na^+	161	O4 (uracil)	511
Ca^{2+}	29	N7 (purine)	447
Mn^{2+}	12	O2' (ribose)	196

Mg^{2+} outnumbered all other cations by roughly 9:1, so an ion-aware model that handles a single species should handle Mg^{2+} . Some 83% of direct coordination is to the two non-bridging phosphate oxygens; the next largest class, G(O6)/G(N7), is the textbook major-groove guanine site.

Coordination is predominantly water-mediated: a mean of 0.77 inner-sphere against 7.00 outer-sphere RNA contacts per Mg^{2+} . Crucially the inner-sphere fraction is itself a function of

tertiary compactness: 0.511 overall but only 0.296 for chains shorter than 500 nt. The inner/outer ratio is therefore a *predictable readout* of compactness, not merely an input.

Stratification warning. These aggregates are ribosome-dominated and must not be quoted flat. The five largest structures hold 39.4% of all Mg^{2+} . Per structure the median is 72.5 ions against a mean of 181.5; per nucleotide the median is 0.046, roughly one Mg^{2+} per 22 nucleotides.

What crystallography cannot see. Resolved Mg^{2+} are site-bound ions only. The diffuse Manning atmosphere, which neutralises roughly 76% of backbone charge, is invisible to crystallography. The two channels must be modelled separately and never conflated.

3.4 Coevolution is strong, but gated by alignment depth

Coevolution is the oldest structural signal in the field: a base pair tolerates mutation at one position only when compensated at the other, so paired columns covary across a family alignment. We measured how much of that signal is actually recoverable, computing sequence-reweighted, pseudocounted, APC-corrected mutual information on the 12 downloaded Rfam seed alignments and scoring it against each family’s curated consensus secondary structure ($\text{\#GC SS_cons}$), restricted to $|i - j| \geq 4$.

Table 6: APC-corrected MI against curated consensus base pairs. **[measured]**

Family	N	N_{eff}	L	$\text{prec}@L/5$	$\text{rec}@L$
tRNA (RF00005)	954	245.9	71	1.000	1.000
THF riboswitch (RF01831)	101	27.9	100	1.000	0.906
Purine riboswitch (RF00167)	133	102.7	102	0.950	0.955
5S rRNA (RF00001)	712	10.3	119	0.826	0.765
RNase P bacterial (RF00010)	458	11.6	367	0.753	0.784
SAM riboswitch (RF00162)	457	19.9	108	0.714	0.697
Cobalamin (RF00174)	434	63.1	189	0.676	0.667
Glutamine (RF01739)	940	15.6	53	0.500	0.692
ydaO/yuaA (RF00379)	106	18.4	134	0.462	0.615
TPP riboswitch (RF00059)	115	41.3	105	0.429	0.926
tmRNA (RF00023)	477	74.6	378	0.400	0.670
FMN riboswitch (RF00050)	146	21.2	139	0.333	0.536
mean				0.670	0.768

The decisive pattern is the depth dependence. Splitting on N_{eff}/L : families at $N_{\text{eff}}/L \geq 1$ ($n = 2$) reach mean $\text{precision}@L/5$ of **0.975**, while those below ($n = 10$) reach 0.609. tRNA, at $N_{\text{eff}}/L = 3.5$, is perfect on both precision and recall.

Note how *few* families are deep: only 2 of 12, and these are hand-curated Rfam seed alignments, among the best-conditioned RNA alignments in existence. Automatic homology search on an arbitrary sequence will generally do worse.

Architectural consequence, and the strongest argument for MoE in this design. Coevolution is not a weak feature to be blended in at fixed weight—where the alignment is deep it is very nearly sufficient on its own. But its availability is conditional, and the condition N_{eff}/L is *cheaply computable before the model runs*. The correct expert therefore depends on an observable property of the input, which is exactly what a mixture of experts is for. N_{eff}/L becomes an explicit router feature, and the coevolution expert is gated on it. This also explains why RhoFold+ pairs a language model *with* MSA features rather than choosing between them: the two have complementary failure modes. PHAROS makes that complementarity routed and explicit instead of concatenated.

An implementation note worth recording. A first version of this measurement reported mean precision 0.039. That was a defect, not a property of RNA: the outer-product term in the MI normalisation broadcast to shape $(C, q, 1)$ rather than (C, q, q) , so only diagonal terms contributed. After correcting it and adding standard DCA sequence reweighting at 80% identity with pseudocounts, precision rose seventeenfold to 0.670. A plausible-looking weak result was a silent bug.

Caveats. The ground truth here is Rfam’s consensus *secondary* structure, not 3D contacts, so this measures recovery of curated pairing rather than direct tertiary contact accuracy. Twelve curated seed families is a small and favourable sample. APC-corrected MI is also a simplification of full DCA; pseudo-likelihood DCA would do better, so these values are a reasonable lower bound on the available signal.

3.5 A negative result that constrains the design

We tested whether GNRA-like tetraloop sequence context alone predicts rigidity, matching the 4-mers {GAAA, GCAA, GAGA, GUGA, GGAA, GCGA, GUAA, GAAG} anywhere in a chain.

Table 7: GNRA k -mer context versus rigidity. **[measured]** Effect size 0.073σ .

Context	n nucleotides	mean z_B
GNRA-like 4-mer	4,944	−0.082
All other	95,768	−0.009

The effect is negligible. The likely cause is methodological: a GNRA *sequence* is a GNRA *tetraloop* only when it actually caps a hairpin, and linear 4-mer matching counts mostly helical and junction occurrences.

Architectural consequence. Sequence context alone does not identify a motif. Motif identity requires the interaction graph. The motif machinery must therefore be conditioned on the pair representation after secondary structure is available, never on raw k -mers. This rules out the naive “learn motifs from sequence n -grams” design.

4 The PHAROS architecture

4.1 Inputs

Input	Required	Note
Sequence, L tokens, vocab 5	yes	elDORS is DNA-alphabet; $T \rightarrow U$ at load
Ionic condition c_{ion}	yes, defaulted	$([\text{Mg}^{2+}], [\text{K}^+], [\text{Na}^+], T, \text{pH})$; novel
MSA / Rfam alignment	optional	graceful degradation when shallow
Chemical probing profile	optional	not yet acquired (Section 6)

Ionic condition enters as a FiLM-style conditioning vector, analogous to a diffusion timestep embedding. The same sequence at 0 mM and 10 mM Mg^{2+} must produce different predictions; this is the clearest novelty claim in the design.

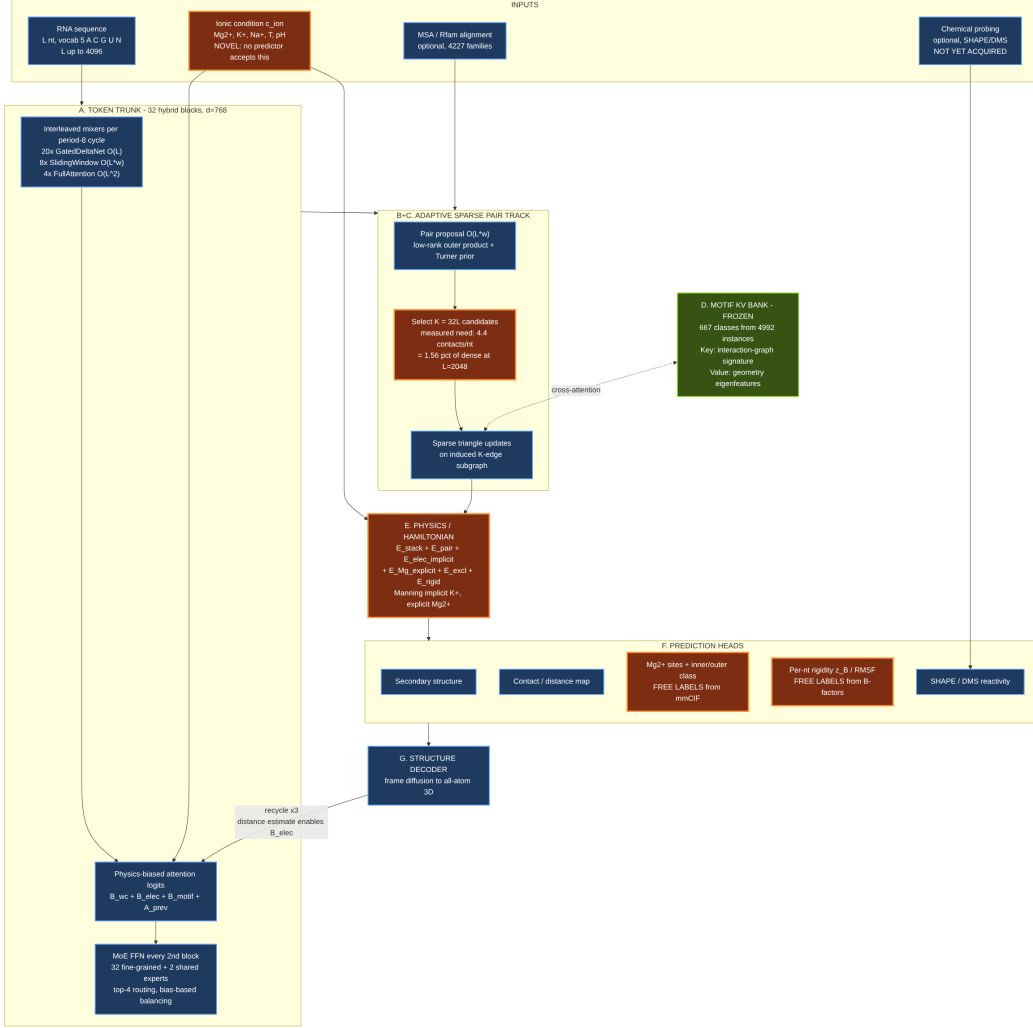


Figure 1: PHAROS system overview. Orange denotes novel components, green denotes frozen retrieval, blue denotes adapted standard components.

4.2 Token trunk: hybrid attention with MoE

The trunk uses 32 blocks at $d = 768$ with the repeating pattern

$$[\text{GDN}, \text{GDN}, \text{SWA}, \text{GDN}, \text{GDN}, \text{SWA}, \text{GDN}, \text{FULL}],$$

where GDN is Gated DeltaNet linear attention ($O(L)$, 20 of 32 blocks), SWA is sliding-window softmax attention with window 128 ($O(Lw)$, 8 blocks), and FULL is full attention with physics bias ($O(L^2)$, only 4 blocks).

The justification is both architectural and biological. Jamba interleaves Mamba, attention and MoE at a 7 : 1 ratio and reaches 256K context with a 4 GB KV cache; Gated DeltaNet outperforms Mamba2 and DeltaNet on in-context retrieval and length extrapolation; Oryx varies the mixer type *within* a sequence. RNA has exactly the structure this exploits: long stretches of regular A-form helix that are locally periodic and low-information, punctuated by rare junctions, pseudoknots and kissing loops that require global mixing. At $L = 4096$, an all-dense 32-block trunk would perform 537M pair evaluations; PHAROS performs 67M, an $8\times$ reduction.

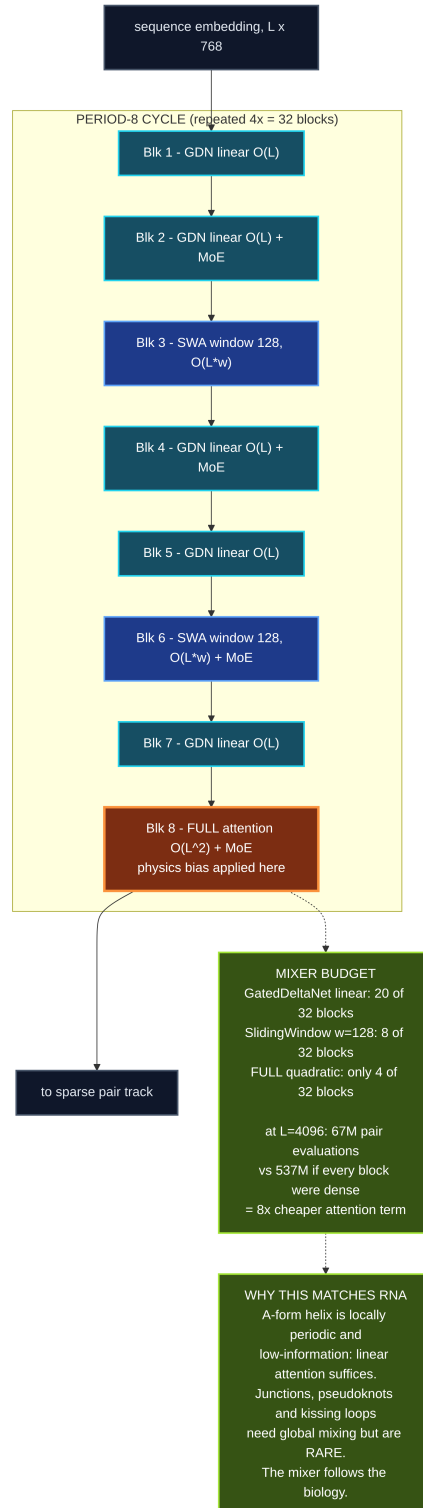


Figure 2: Period-8 mixer interleaving, repeated four times for 32 blocks.

4.2.1 Physics-biased attention

In SWA and FULL blocks the attention logits carry four additive terms:

$$A_{ij} = \frac{Q_i K_j^\top}{\sqrt{d}} + B_{\text{wc}}(i, j) + B_{\text{elec}}(i, j; c_{\text{ion}}) + B_{\text{motif}}(i, j) + A_{ij}^{(l-1)}. \quad (2)$$

B_{wc} generalises ERNIE-RNA’s fixed prior to a learned pair-type embedding modulated by stacking context, since real pairing energy is nearest-neighbour dependent. It is initialised at the ERNIE-RNA values so training begins from a known-good prior. The term $A^{(l-1)}$ retains ERNIE-RNA’s attention accumulation.

B_{elec} is the novel term. With Bjerrum length $\ell_B = e^2/(4\pi\epsilon_0\epsilon_r k_B T) \approx 7.1 \text{ \AA}$ in water at 298 K, Manning parameter $\xi = \ell_B/b$ for axial charge spacing b , condensed fraction and renormalised charge

$$\theta = 1 - \frac{1}{z\xi}, \quad q_{\text{eff}} = -(1 - \theta), \quad (3)$$

and Debye screening parameter $\kappa = \sqrt{2N_A e^2 I / (\epsilon\epsilon_0 k_B T)}$ from ionic strength I , the bias is

$$B_{\text{elec}}(i, j) = -\lambda q_{\text{eff}}^2 \ell_B \frac{\exp(-\kappa d_{ij})}{d_{ij}}. \quad (4)$$

For monovalent RNA $\theta \approx 0.76$, so roughly three quarters of backbone charge is screened before any structure-specific electrostatics is considered. Only λ is learned; q_{eff} , κ and ℓ_B follow in closed form from c_{ion} and temperature. Equation (4) is the mechanism by which salt concentration changes the prediction.

Resolving the circularity. Equation (4) requires d_{ij} , which is unknown on the first pass. B_{elec} is therefore disabled at recycle 0 and enabled from recycle 1 using the current predicted distance map. Recycling is thus structural to the design rather than an optimisation: the physics term needs a geometry estimate to evaluate, and each recycle sharpens both.

4.2.2 Mixture-of-experts feed-forward

Following DeepSeekMoE, the design uses 32 fine-grained routed experts at $d_{\text{ff}} = 512$ plus 2 always-on shared experts, with top-4 routing giving 6 active experts of 34, in every second block.

Shared-expert isolation matters for a specific reason here: Watson–Crick helix formation is the highest-frequency pattern in all RNA. Placing it in an always-on shared expert prevents all 32 routed experts from separately relearning it, freeing them to specialise on what is rare and hard—junctions, pseudoknots and ion pockets.

Load balancing is auxiliary-loss-free, adjusting a per-expert bias each step rather than adding a loss term. This matters acutely here because our 3D data is pathologically imbalanced: 5S rRNA alone accounts for 5,976 BGSU clusters, so an auxiliary balance loss would actively fight the biology.

The router input is the token hidden state concatenated with the current local structural estimate—pairing probability, predicted rigidity, local density—and the alignment depth N_{eff}/L , which gates the coevolution expert (Section 3.4). This follows directly from the GNRA negative result of Table 7: routing on sequence k -mers does not work. It also prevents router collapse onto rRNA, because structural regimes are far more evenly distributed across the corpus than families are.

4.2.3 Parameter budget

PHAROS-Base carries $2.3\times$ the capacity of NucleicBERT at slightly lower active compute, with an attention term roughly $8\times$ cheaper at long context.

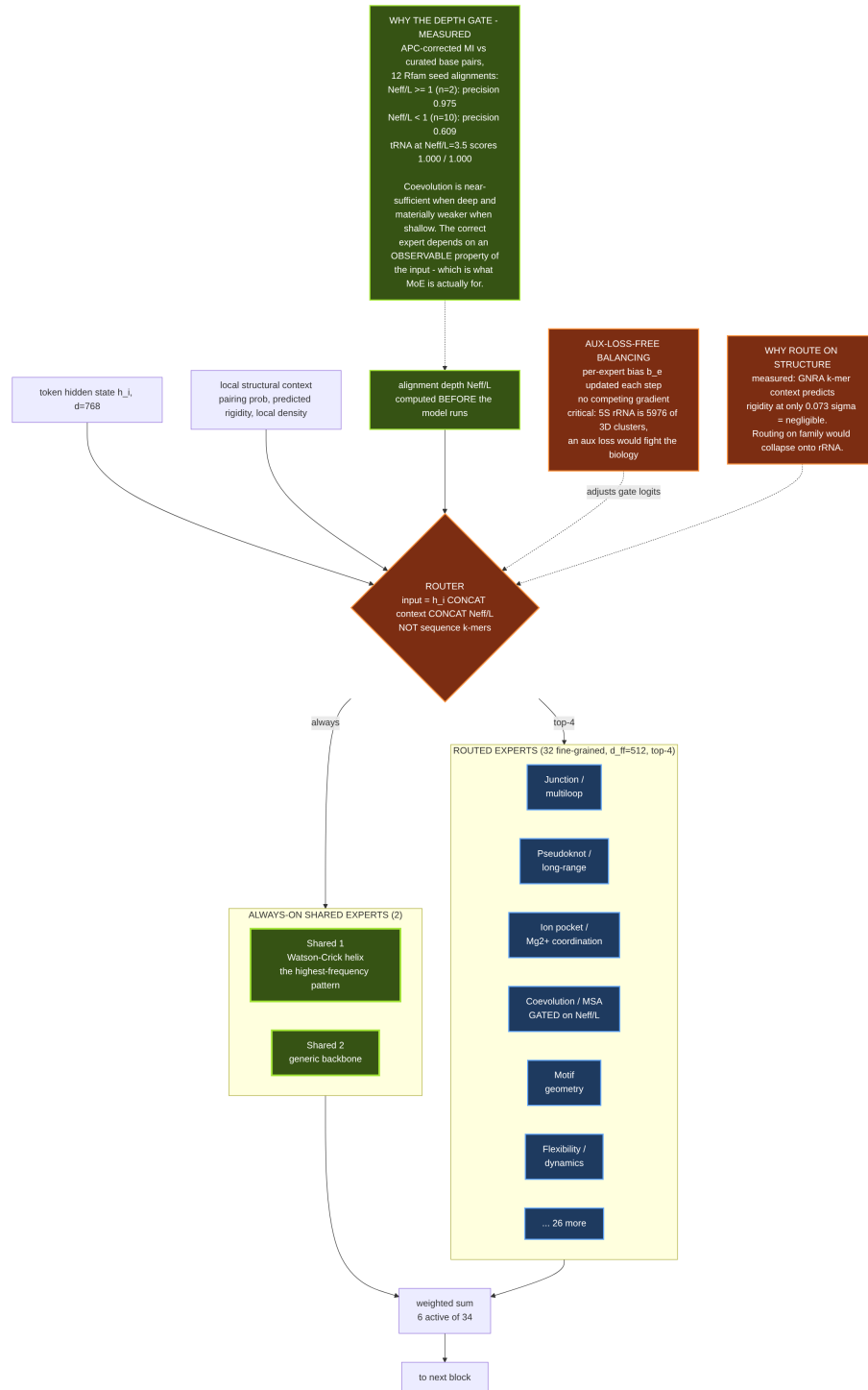


Figure 3: Expert routing. Routing keys on local structural regime, not sequence.

Table 8: PHAROS-Base parameter budget against NucleicBERT.

Component	Total	Active
Embeddings (vocab 5)	0.01M	0.01M
Attention, 32 blocks, $d = 768$	75.5M	75.5M
MoE FFN, 16 blocks $\times$ 34 experts	642M	113M
Dense FFN, 16 non-MoE blocks	113M	113M
Sparse pair track and triangle updates	45M	45M
Motif bank, heads, decoder	35M	35M
PHAROS-Base total	$\sim 911\text{M}$	$\sim 382\text{M}$
NucleicBERT (dense)	404M	404M

4.3 Hierarchical pair track

4.3.1 A flat top- K proposal was specified, measured, and discarded

Version 0.1 of this design specified a flat proposal retaining $K = 32L$ candidate edges. We tested it. Scoring every pair with a sequence-only proposer (the ERNIE-RNA pair prior, stacking context, and an exponential sequence-distance decay) and measuring recall of true 3D contacts within the top K gives Table 9.

Table 9: Recall of true 3D contacts under a flat top- K proposal at $c = 32$. **[measured]**

Chain length	n	mean recall	K as % of dense
32–100	29	0.980	59.4%
100–200	6	0.642	25.5%
200–500	3	0.307	9.0%
500–1200	5	0.200	4.7%

A *random* scorer achieves mean recall 0.746 at $c = 32$ over the same chains, because on short chains $32L$ already covers most of the map. The flat proposer’s headline mean of 0.795 is therefore very nearly an artefact of short chains. **On long chains, where sparsity actually matters, it recovers one contact in five**—a silent and fatal accuracy ceiling.

A banded fallback does not rescue it, because RNA contacts are not sufficiently local. The fraction of contacts within $|i - j| \leq \text{band}$ is 0.350 at band 32, 0.479 at 64, 0.597 at 128, 0.706 at 256 and 0.812 at 512 for chains of 1500–3000 nt; median contact separation is 78 nt with a long tail. Even a ± 512 band, which would cost vastly more than the whole budget, misses 19%.

4.3.2 Contacts are clustered, so select blocks rather than pairs

Helical stems are contiguous anti-diagonal runs in the contact map. Contacts are therefore spread along the sequence but concentrated in very few *blocks*. Table 10 measures this.

Table 10: Block occupancy at block size $b = 4$: the fraction of blocks containing at least one contact, and the effective c of keeping every occupied block. **[measured]**

Chain length	median L	occupancy at $b = 4$	effective c
64–200	88	16.34%	7.9
200–500	393	6.87%	12.6
500–1500	1038	3.17%	14.8
1500–3000	2861	1.34%	17.2

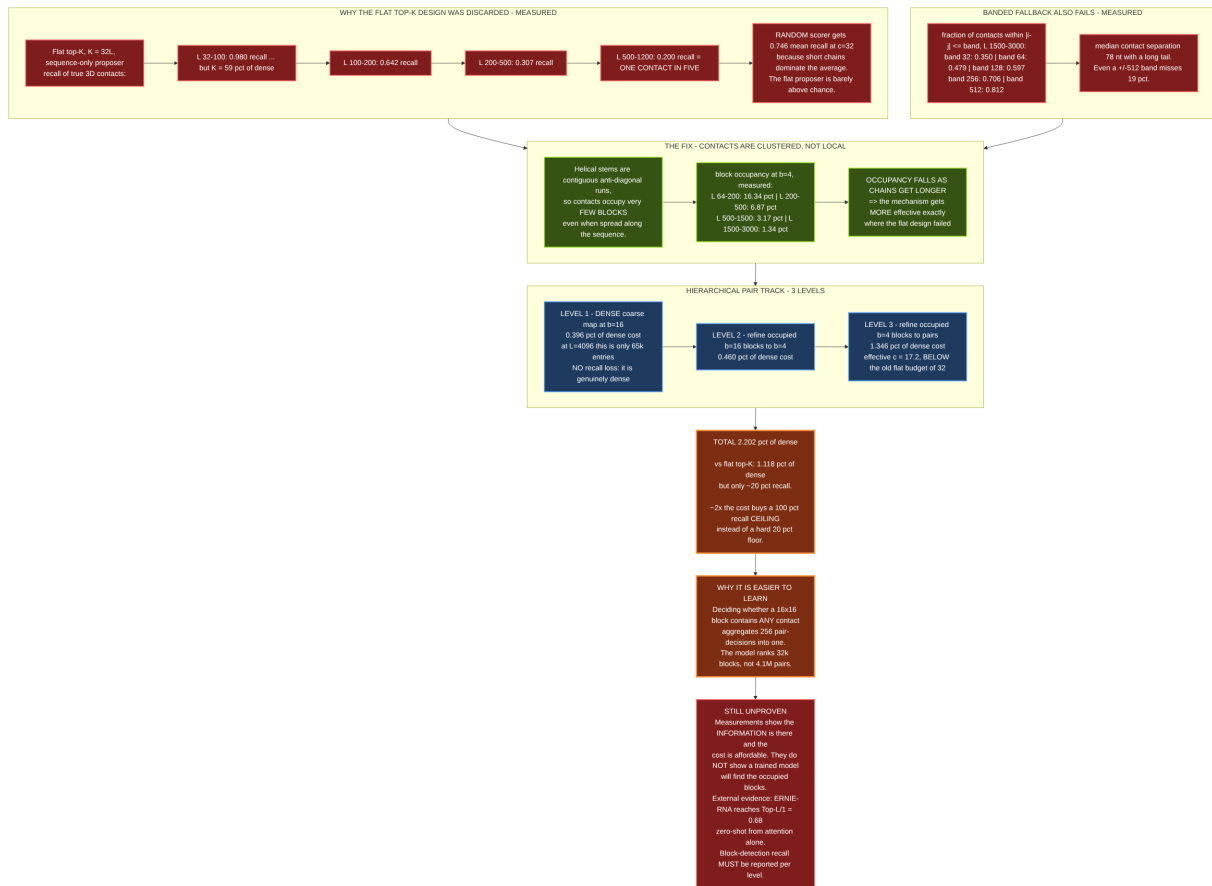


Figure 4: The pair track, after the flat design was measured and replaced.

Occupancy *falls* as chains lengthen, so the mechanism becomes more effective at exactly the lengths where the flat design failed. Retaining every occupied 4×4 block costs an effective c of roughly 17—below the discarded flat budget of 32—while capturing every contact by construction.

4.3.3 Three-level coarse-to-fine track

Table 11: Cost of the hierarchical track at $L = 2861$, as a fraction of a dense pair map. [measured]

Level	Operation	Cost
L1	dense pair map at block size $b = 16$	0.396%
L2	refine occupied $b = 16$ blocks to $b = 4$	0.460%
L3	refine occupied $b = 4$ blocks to individual pairs	1.346%
Total		2.202%
flat top- K at $c = 32$, for comparison		1.118% (at $\sim 20\%$ recall)

For roughly twice the cost of the flat design, the track moves from a hard 20% recall ceiling to a ceiling of 100%. Two properties make this work. First, the L1 map is *genuinely dense*, so it loses no recall at the coarse level at all—at $L = 4096$ with $b = 16$ it is only 65k entries. Recall can only be lost at the selection thresholds, which are tunable and can be set recall-first. Second, deciding whether a 16×16 block contains any contact aggregates 256 pair-decisions into one, so coarse-level signal-to-noise is far higher than for individual pair ranking. The model ranks 32k blocks and then 16 sub-blocks inside each survivor, never 4.1M individual pairs.

4.3.4 Reference implementation and measured cost

The cost arithmetic above is turned into running code in `research/architecture/reference/`, and measured against an AlphaFold3-style dense baseline on the machine this report was prepared on (CPU only, 14 GB RAM, no CUDA), at batch 1, $d_{\text{model}} = 768$, $d_{\text{pair}} = 128$, $b_1 = 16$, $b_2 = 4$ and budget $c = 20$.

Table 12: Hierarchical track versus dense baseline, measured. [measured]

L	HPT time	HPT pairs	% of dense	eff. c	Dense time	Dense peak RSS
128	0.01 s	750	9.23%	5.9	0.07 s	0.10 GB
256	0.01 s	3,506	10.74%	13.7	0.28 s	0.33 GB
512	0.05 s	10,054	7.69%	19.6	1.22 s	1.30 GB
1024	0.10 s	20,102	3.84%	19.6	8.26 s	5.15 GB
2048	0.33 s	40,198	1.92%	19.6	<i>not runnable</i>	predicted 12.9 GB
4096	0.45 s	80,390	0.96%	19.6	<i>not runnable</i>	predicted 51.5 GB

Two things are established that arithmetic alone could not. First, the dense baseline genuinely does not run: it needs a predicted 12.9 GB of activations at $L = 2048$ and 51.5 GB at $L = 4096$ on a 14 GB machine, so the infeasibility claim is not rhetorical. Second, at the largest length both can run, the hierarchical track is roughly $83\times$ faster (0.10 s versus 8.26 s) with peak resident memory below measurement noise against the baseline’s 5.15 GB.

The cost fraction falls monotonically with length, from 9.2% to 0.96%, because the budget grows as cL while the dense map grows as L^2 . The mechanism becomes cheaper relative to dense exactly where it is needed, mirroring the falling block occupancy of Table 10.

Two silent defects found while building it. Both would have degraded a trained model rather than crashing. The L3 expansion initially paired block rows with block columns element-wise, materialising only each block’s *diagonal*—4 pairs per 4×4 block instead of 16—silently

delivering a quarter of the intended budget. Separately, the L2 selection budget was expressed as a fraction of all pairs and was clamped by masking to a quarter of its target. Both pinned the effective c at 5.0 while the code appeared to work. Budgets are now stated in the same units as the measurement ($K = cL$ pairs) so any mismatch is immediately visible.

What remains unproven. These measurements establish that the information is present and the cost affordable. They do *not* establish that a trained model will identify occupied blocks accurately; that requires training. The relevant external evidence is ERNIE-RNA’s zero-shot Top- $L/1$ contact precision of 0.68 from attention maps alone, which indicates that learned representations rank contacts far better than the sequence-only heuristic tested above. **Block-detection recall must be reported per length bin and per level.**

4.4 Motif retrieval bank

The RNA 3D Motif Atlas 4.12 compresses 4,992 loop instances into 667 motif classes, 413 internal and 254 hairpin: a 7.5 : 1 ratio indicating that RNA tertiary structure is far more repetitive than its sequence diversity suggests. PHAROS precomputes a frozen key–value bank over these classes, with keys encoding the non-canonical interaction-graph signature and values encoding internal geometry as distance-matrix eigenfeatures plus a backbone torsion profile. The pair track cross-attends into this bank.

This is the appropriate response to data scarcity. With only 6,661 unique sequences carrying 3D structure, a model cannot reliably learn motif geometry implicitly. The bank supplies the geometry explicitly, leaving the model to learn only *where* motifs occur—a substantially easier problem than learning what they look like. The bank is roughly 10^3 fixed entries shared across a batch, so its cost is negligible. Consistent with Table 7, retrieval is keyed on the interaction graph, never on sequence k -mers.

4.5 Physics module

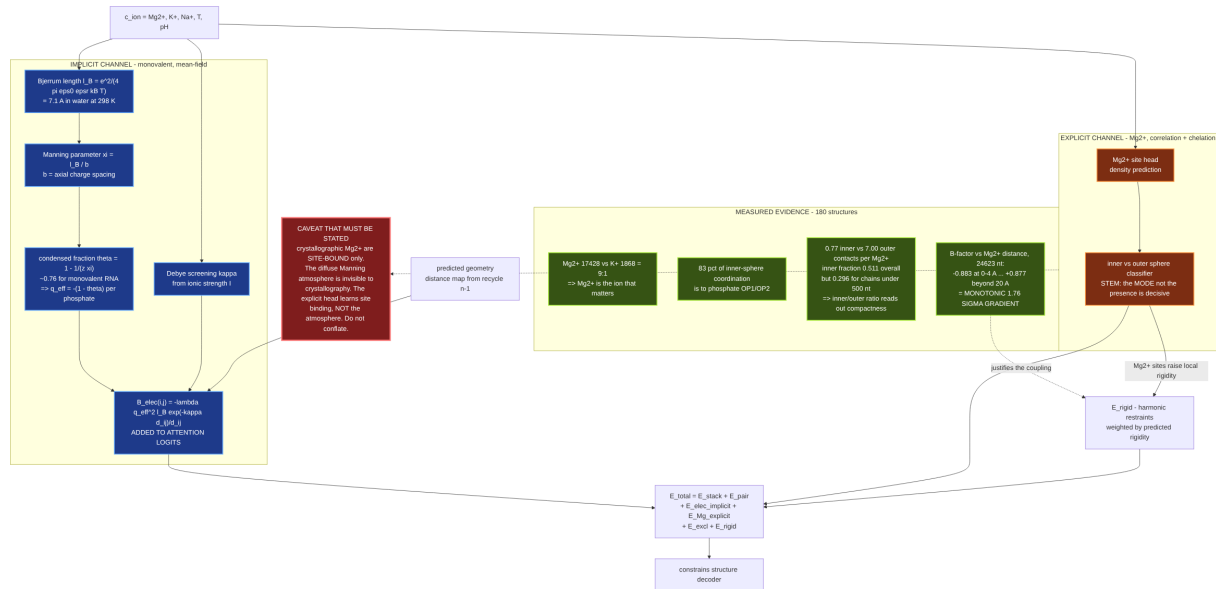


Figure 5: The hybrid implicit/explicit electrostatic treatment.

A differentiable energy is evaluated on the predicted geometry:

$$E_{total} = E_{stack} + E_{pair} + E_{elec}^{implicit} + E_{Mg}^{explicit} + E_{excl} + E_{rigid}. \quad (5)$$

The implicit/explicit split follows STEM, which establishes that mean-field theory captures monovalent KCl well but fails for Mg^{2+} , where ion–ion correlation and site-specific chelation dominate. Our own inventory agrees on which species matters: Mg^{2+} outnumbers K^+ by 9:1.

Inner/outer-sphere classification is a first-class output because STEM’s central claim is that folding dynamics are driven by *exchange between* direct and water-mediated coordination—the mode, not the presence, is decisive. Our measurements supply both channels directly and show the inner fraction varies with compactness (0.511 overall, 0.296 below 500 nt).

The term E_{rigid} closes the loop with Table 4: predicted Mg^{2+} sites raise local rigidity, which stiffens harmonic restraints, which constrains the decoder. That pathway is the architectural encoding of the measured 1.76σ gradient.

4.6 Heads and decoder

Table 13: Prediction heads and their supervision. Two channels are effectively free.

Head	Supervision source	Status	Labels
Secondary structure	bpRNA, ArchiveII, RNAStrAlign	in catalogue	~160k
Contact / distance	RNA3DB, gRNAde	in catalogue	27,452 chains
Mg^{2+} sites	extracted from mmCIF	free	17,428 (180 structs)
Rigidity (z_B , RMSF)	B-factors in mmCIF	free	every X-ray struct
Reactivity (SHAPE/DMS)	Ribonanza / RMDB	not acquired	—
3D coordinates	RNA3DB	in catalogue	6,661 unique seqs

The decoder is frame-based diffusion. We retain an explicit geometric and energetic stage rather than going fully generative, consistent with the CASP16 observation that physics-plus-ML hybrids currently lead in RNA and with trRosettaRNA’s restraint-then-fold approach outscoring AlphaFold3’s raw-atom diffusion.

Mg^{2+} sites and B-factor rigidity deserve emphasis: both are already present inside every mmCIF file the field routinely downloads, are used by no current structure predictor, and were extracted end-to-end in preparing this report.

5 Ionic-condition supervision: an audit that rescoped the claim

The headline novelty is accepting ionic conditions as an input. That requires structures labelled with the ionic environment they were solved in, so we audited all 180 sampled structures. The result changed the claim.

5.1 Coverage is roughly one third, split across two storage conventions

Table 14: Where ionic conditions live, and how often they are populated. **[measured]**

Method	n	mmCIF location	usable ionic vector
Cryo-EM	117	<code>_em_buffer_component</code> (structured)	23.9%
X-ray	63	<code>_exptl_crystal_grow.pdbx_details</code> (free text)	58.7% mention an ion

Cryo-EM entries record pH for 94% of structures but populate `_em_buffer_component` for only 24.8%. X-ray entries always carry crystallisation details, but as prose—“0.1–0.2 M Arginine-HCl, 0.1M Tris-HCl pH 7.6, 2.6–3.0% PEG-20K”—so extraction is noisy. Combined, about 36% of structures yield any recoverable ionic condition: trainable, but a threefold reduction in usable 3D data for that stage.

5.2 The deeper problem: the PDB is survivorship-biased on ionic conditions

Where Mg^{2+} concentration *is* recorded, its range is narrow.

Table 15: Recorded buffer concentrations in cryo-EM entries. **[measured]**

Ion	n	p10	median	p90
Mg^{2+}	21	5.0 mM	7.5 mM	15.0 mM
K^+	13	30 mM	100 mM	150 mM
Na^+	12	100 mM	150 mM	150 mM

Mg^{2+} spans only 5–15 mM. This is not a sampling accident: **nobody deposits a structure of unfolded RNA**. Every PDB entry was solved under conditions chosen *because* the RNA folds there, so the archive contains almost no contrast between folding and non-folding ionic regimes.

Consequence. The $[\text{Mg}^{2+}] \rightarrow \text{structure}$ response cannot be learned from the PDB. A model trained only on deposited structures would observe ionic condition as a near-constant and would correctly learn to ignore it.

5.3 How the claim is rescoped

The novelty splits into three components with very different evidentiary support.

Component	Support	Status
B_{elec} Manning/Debye screening	closed-form physics; needs no training labels	unaffected
Mg^{2+} site head and inner/outer classifier	17,428 labelled ions from 180 structures	well supported
Learned <i>response</i> to varying ionic conditions	requires titration series; the PDB has none	not supported

The first two stand unchanged. The third must be dropped or supplied from data the PDB cannot provide: **RMDB Mg^{2+} titration series** (SHAPE/DMS probing across an Mg^{2+} ladder on the same sequence) and SAXS Mg^{2+} titrations measuring compaction. Training stage 5 as originally specified—ion-conditioned refinement on structures stratified by ionic condition—is **not viable on PDB data alone** and is rewritten to condition on probing titrations, with deposited structures supplying only site-level supervision.

This promotes chemical-probing acquisition from “most valuable missing asset” to **prerequisite for the headline claim**. Note that Ribonanza alone does not supply it; the Mg^{2+} titration series in RMDB do.

6 Data mapping and the acquisition gap

Six of seven training stages are fully served by the existing catalogue. The exception is chemical probing, and it is the most valuable missing asset. Ribonanza provides 2.1M sequences with DMS and SHAPE measurements—roughly $315\times$ more supervised examples than the 6,661 unique sequences with 3D structure. It is the only label source that is simultaneously abundant, structure-informative and already public. This is the highest-priority acquisition.

Corpus weighting. The eLDORS chunks are source-partitioned rather than homogeneous: 9 of 20 chunks have a median length of exactly 151 nt, indicating unassembled sequencing reads, while chunks 6 and 9 (median 2,193–2,389 nt) are the only substantial source of long-context signal.

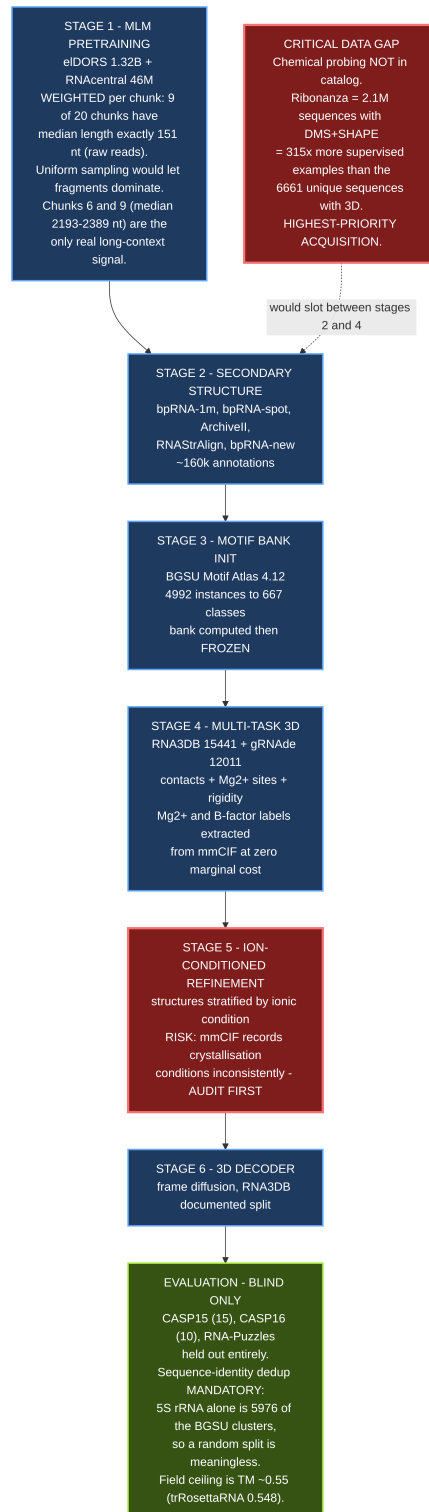


Figure 6: Training curriculum, data sources, and the chemical-probing gap.

Uniform sampling in Stage 1 would allow read fragments to dominate pretraining, so per-chunk weighting is required.

7 Novelty claims

Table 16: Novelty claims stated precisely, with prior-art status.

Claim	Prior-art status
Ionic conditions as a model input, reaching attention logits through a Manning-screened Coulomb bias	No existing RNA structure predictor accepts ionic conditions. The physics term needs no labels; the learned response requires titration data (Section 5).
First sparse-MoE architecture for RNA structure	AIDO.Protein is MoE for protein; no RNA equivalent exists.
Hierarchical coarse-to-fine pair track selecting blocks rather than pairs, justified by measured 1.34% block occupancy	AlphaFold3 and Rhoformer are dense $O(L^2)$; flat top- K sparsification is measured here to fail at $\sim 20\%$ recall on long chains.
Frozen motif KV bank as a retrieval-based geometry prior	The Motif Atlas is published but wired into no large model.
Mg ²⁺ sites and B-factor rigidity as free auxiliary supervision	Present in mmCIF; used by no structure predictor.
Coupled ion-rigidity expert justified by a measured 1.76σ gradient	Treated separately or not at all elsewhere.

8 Risks and open questions

1. **Block-detection recall.** Superseded but not eliminated. The flat top- K design was measured at $\sim 20\%$ recall on long chains and replaced (Section 4.3). The hierarchical track has a 100% recall *ceiling* by construction, but whether a trained model attains it is unproven. Recall must be reported per length bin and per level.
2. **Ionic metadata sparsity—audited, partially confirmed** (Section 5). About 36% of structures yield a recoverable ionic condition. Worse, recorded Mg²⁺ spans only 5–15 mM because the PDB is survivorship-biased. The learned ionic response must come from RMDB titration series, not the PDB. The closed-form physics term and the Mg²⁺ site head are unaffected.
3. **Crystallographic ions are not the solution ensemble.** The explicit head learns site binding, not the diffuse atmosphere. These must not be conflated.
4. **B-factor is a noisy rigidity proxy.** Excluding cryo-EM left only 31 of 137 parsed structures usable, which materially shrinks the rigidity label set.
5. **Association, not isolated causation.** Mg²⁺ proximity and packing density both track folded cores.
6. **Infrastructure.** NucleicBERT used 192 A100s densely. MoE improves FLOP efficiency but adds routing and communication complexity.
7. **Evaluation honesty.** The field ceiling is TM ≈ 0.55 . Any improvement claim must be made on blind sets—CASP15/16 and RNA-Puzzles—never on rRNA-saturated random splits. Sequence-identity deduplication is mandatory.

9 Status

This is a design specification. No model has been trained and no predictive performance is claimed. What has been established is the evidentiary basis: the contact-scaling law, the ion-rigidity coupling, the ion inventory and coordination chemistry, and the GNRA negative result, all measured on 180 non-redundant structures with reproducible scripts. The next phase is to implement the sparse pair track and validate proposal recall before any large-scale training is attempted.

A Reproducing the measurements

Script	Output
.../sampling/fetch_samples.py	180 mmCIF, 12 Rfam seeds
.../sampling/analyze_ions_motifs.py	ion_summary.json
.../sampling/analyze_rigidity.py	rigidity_summary.json
.../sampling/analyze_contact_sparsity.py	contact_sparsity.json
.../sampling/validate_proposal_recall.py	proposal_recall.json
.../sampling/analyze_contact_separation.py	contact_separation.json
.../sampling/analyze_block_sparsity.py	block_sparsity.json
.../sampling/audit_ionic_metadata.py	ionic_metadata_audit.json
.../sampling/audit_em_buffers.py	em_buffer_audit.json
.../sampling/measure_coevolution.py	coevolution_signal.json
.../reference/benchmark_pair_track.py	pair_track_benchmark.json

References

- [1] ERNIE-RNA: an RNA language model with structure-enhanced representations. *Nature Communications*, 2025. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12627772/>
- [2] Accurate RNA 3D structure prediction using a language model-based deep learning approach (RhoFold+). *Nature Methods*, 2024. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11621015/>
- [3] Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 2024. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11168924/>
- [4] Structural Topology-based Electrostatic Model (STEM) Reveals Ion-Coordination Exchange as a Driver of RNA Folding Dynamics, 2026. <https://pmc.ncbi.nlm.nih.gov/articles/PMC13345221/>
- [5] Generalized Manning Condensation Model Captures the RNA Ion Atmosphere, 2015. <https://pubmed.ncbi.nlm.nih.gov/26197147/>
- [6] The RNA 3D Motif Atlas: Computational methods for extraction, organization and evaluation of RNA motifs. *Methods*, 2016. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4921307/>
- [7] Ribonanza: deep learning of RNA structure through dual crowdsourcing, 2024. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10925082/>
- [8] DeepSeekMoE: Towards Ultimate Expert Specialization in Mixture-of-Experts Language Models, 2024. <https://arxiv.org/html/2401.06066v1>
- [9] DeepSeek-V3 Technical Report, 2024. <https://arxiv.org/pdf/2412.19437>
- [10] Gated Delta Networks: Improving Mamba2 with Delta Rule. *ICLR*, 2025. <https://arxiv.org/pdf/2412.06464>
- [11] Mixture of Experts Enable Efficient and Effective Protein Understanding and Design (AIDO.Protein). *bioRxiv*, 2024. <https://www.biorxiv.org/content/10.1101/2024.11.29.625425v1>

-
- [12] Deep learning for RNA structure prediction. *Current Opinion in Structural Biology*, 2025. <https://www.sciencedirect.com/science/article/pii/S0959440X25000090>
 - [13] Enhancing RNA 3D Structure Prediction in CASP16: Integrating Physics-Based Modeling with Machine Learning, 2025. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12354339/>
 - [14] Elastic network models for RNA: a comparative assessment with molecular dynamics and SHAPE experiments. *NAR*, 2015. <https://academic.oup.com/nar/article/43/15/7260/2414435>
 - [15] Graph deep learning locates magnesium ions in RNA, 2023. <https://pubmed.ncbi.nlm.nih.gov/37292390/>
 - [16] Comprehensive benchmarking of large language models for RNA secondary structure prediction. *Briefings in Bioinformatics*, 2025. <https://academic.oup.com/bib/article/26/2/bbaf137/8109668>